Module Quiz and Explanation

NumPy MCQs for Beginners (Part 2 Topics)

1) What will be the shape of the array in the output?

```
import numpy as np
a = np.array([1, 2, 3, 4, 5, 6])
b = a.reshape(2, 3)
print(b.shape)
```

A. (3, 2)

B. (2, 3)

C. (6,)

D. (3, 3)

✔ **Correct Answer:** B
Explanation: reshape(2,3) মানে 2 টা row আর 3 টা column। তাই উত্তর (2,3)।

2) What is the result of np.sin(np.pi/2)?

A. 0

B. 0.5

C. 1.0

D. -1.0

✔ **Correct Answer:** C
Explanation: π/2 রেডিয়ান মানে 90 ডিগ্রি, sin(90°)=1। তাই উত্তর 1.0।

3) What will be the output of the following code?

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(a + b)
```

A. [5 7 9]

B. [14]

C. [4 5 6 1 2 3]

D. Error

✔ **Correct Answer:** A
Explanation: NumPy তে array যোগ মানে element-wise যোগ হয়। তাই [1+4, 2+5, 3+6] = [5 7 9]।

4) Which function returns the standard deviation of an array?

A. np.mean()

B. np.std()

C. np.var()

D. np.average()

✔ **Correct Answer:** B
Explanation: np.std() ফাংশনটি array-এর মানের বিচ্যুতি বা standard deviation হিসাব করে।

5) What will be the output of this code?

```
a = np.array([3, 1, 2])
print(np.sort(a))
```

A. [3 1 2]

B. [1 2 3]

C. [2 3 1]

D. Error

✔ **Correct Answer:** B
Explanation: np.sort() array-কে ছোট থেকে বড় ক্রমে সাজায়। তাই [1 2 3]।

6) Given two NumPy matrices, A = np.array([[1, 2], [3, 4]]) and B = np.array([[5, 6], [7, 8]]), what is the correct result of the matrix multiplication dot(A,B)

A. [[5, 12], [21, 32]]
B. [[13, 17], [21, 25]]
C. [[19 22] [43 50]]
D. 64

✔ **Correct Answer:** C

Explanation : এখানে দুটি numpy array এর dot product বের করা হয়েছে।

7) What is NumPy primarily used for in Python?

A. Building web applications

B. Numerical and matrix computations

C. File handling and I/O

D. GUI design

✔ **Correct Answer:** B
Explanation: NumPy মূলত numerical computation, matrix, ও multidimensional array নিয়ে কাজ করে।

8) Which of the following statements about NumPy arrays is true?

A. They are slower than Python lists

B. They consume more memory

C. They are faster and more efficient than lists

D. They can hold mixed data types

✔ **Correct Answer:** C
Explanation: NumPy array গুলো Python list থেকে দ্রুততর ও কম মেমোরি ব্যবহার করে।